



Master in Quantum Technologies & Engineering

FINAL THESIS

Noisy quantum key distribution:

Information leakage of the BB84 protocol

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Academic course 2025-2026

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Noisy quantum key distribution: Information leakage of the BB84 protocol

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(Dated: June 2026)

ABSTRACT

This thesis studies information leakage in the BB84 quantum key distribution protocol under realistic detection constraints and noisy conditions. Rather than restricting the analysis to the full recovery of the secret key, the eavesdropper is modeled as an adversary interested in gaining any useful information about the transmitted data. The framework is based on worst-case leakage metrics related to Sibson mutual information and gentle quantum leakage, where information acquisition is constrained by the need to avoid detection. Special attention is paid to depolarizing noise and to the relation between disturbance parameters, bit error rate, and information leakage. Numerical experiments are proposed to map the trade-off between disturbance and leakage both in ideal and noisy regimes, providing a practical interpretation of leakage bounds in the language of quantum key distribution.

Keywords: BB84, quantum key distribution, Sibson mutual information, maximal leakage, gentle quantum leakage, asymmetric quantum cloning, depolarizing noise, QBER.

I. INTRODUCTION

Quantum Key Distribution (QKD) allows two legitimate parties, Alice and Bob, to establish a shared secret by exploiting fundamental properties of quantum mechanics. Among QKD protocols, BB84 remains the canonical example due to its conceptual simplicity and its foundational role in the theory of quantum cryptography [1].

The usual security intuition behind BB84 is based on the non-orthogonality of the transmitted quantum states. Because the states cannot be perfectly distinguished, any attempt by an eavesdropper to extract information necessarily introduces disturbance. In practice, this disturbance may be detected through an increase in the quantum bit error rate, which makes BB84 a natural setting to study the information–disturbance trade-off [2], [3].

However, traditional security analyses are often centered on the recovery of the full secret key or on average information measures. In many realistic adversarial scenarios, this viewpoint may be too restrictive. An adversary may benefit from learning only partial information, a function of the secret, or any correlated attribute that improves inference. This motivates the use of worst-case leakage metrics, which quantify the maximum advantage available to an eavesdropper [4].

The present thesis studies BB84 under this broader perspective. The goal is to analyze how much information Eve can extract while remaining difficult to detect, and how this leakage changes in the presence of noise. In addition, special attention is given to the effect of depolarizing noise and to the operational re-

lation between leakage, disturbance, and the bit error rate. Fig. 1 shows the eavesdropping scenario considered in this work, where Alice’s transmitted quantum message is affected by depolarizing noise before being intercepted by Eve and finally reaching Bob.

II. BB84 AND LEAKAGE FRAMEWORK

A. The BB84 Protocol

The BB84 protocol [5] uses four pure qubit states distributed over two mutually unbiased bases: in the computational basis,

$$|0\rangle, \quad |1\rangle,$$

and in the diagonal basis,

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

Alice selects two random bit strings for both a bit value and a basis to encode the information into the qubits that she will send to Bob. Bob measures either on the computational or diagonal basis (chosen at random for each qubit). After the quantum transmission stage, both parties publicly compare the bases used and retain only the events where the preparation and measurement bases coincide. We denote this process as sifting.

An essential feature of BB84 is that a measurement generally modifies the quantum state being measured. In the projective-measurement picture, this is described as the collapse of the state onto an eigenstate of the measurement basis. Therefore, if Eve in-

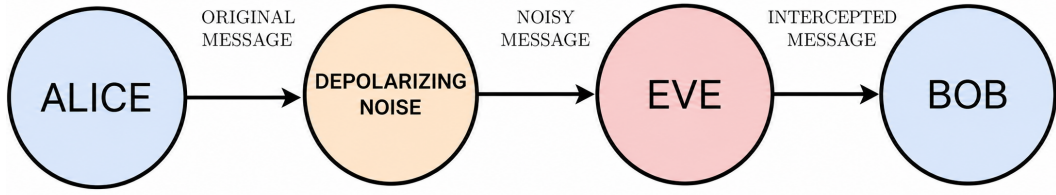


Figure 1. QKD eavesdropping scenario.

tercepts and measures a qubit, she must choose a basis. When her basis differs from Alice’s preparation basis, the state forwarded to Bob is disturbed, and Bob may obtain a bit value different from Alice’s even when he later measures in the correct basis. This disturbance appears as an increased quantum bit error rate (QBER), defined as the probability that Alice and Bob obtain different bits conditioned on having used the same basis:

$$\text{QBER} = \Pr(A \neq B \mid \text{same basis}).$$

The QBER is therefore the main observable quantity used by Alice and Bob to detect noise or a possible eavesdropping attack [1].

After basis sifting, to estimate the quality of the transmission, Alice and Bob publicly reveal a randomly chosen subset of the remaining bits and compare them. These revealed bits are discarded from the raw key. If the estimated QBER is above the tolerated threshold, they abort the protocol; otherwise, they proceed with the classical post-processing stage.

The security intuition of BB84 follows from the impossibility of perfectly distinguishing non-orthogonal states. Since the states from different bases are not mutually orthogonal, an eavesdropper cannot extract full information without introducing disturbance. This is the core mechanism that links information gain to detectability [2], [3].

B. Threat Model and Adversarial Goal

Let X denote Alice’s classical random variable, with finite alphabet $\mathbb{X}$, and assume that $p_X(x) > 0$ for all $x \in \mathbb{X}$. Alice encodes each value x into a quantum state $\rho^x \in \mathcal{S}(\mathcal{H})$. The corresponding ensemble is

$$\mathcal{E} = \{p_X(x), \rho^x\}_{x \in \mathbb{X}}.$$

In this work, Eve is not trying to recover the full raw key X . Instead, she attempts to infer an arbitrary, possibly randomized, discrete function Z of X . This includes bits, subsets of bits, functions of the raw string, or other partial attributes. Specifically, after applying a measurement, Eve observes a classical outcome Y and produces a one-shot estimate $\hat{Z}$ of Z . Alice and Bob do not know the specific target of Eve’s attack.

This motivates the use of worst-case leakage metrics, which capture the largest multiplicative improvement in Eve’s probability of guessing any target variable derived from the secret. A second feature of the threat model is that Eve aims to remain undetected. Therefore, her measurement strategy is not unconstrained: it must satisfy a disturbance requirement. This naturally leads to the notion of gentle measurements and, more generally, gentle quantum leakage [3].

C. Measurement Model

Generalized quantum measurements are modeled using the Positive Operator Valued Measure (POVM) framework [6]. A POVM is a collection of positive semidefinite operators $\{F_y\}_{y \in \mathbb{Y}}$ satisfying

$$\sum_{y \in \mathbb{Y}} F_y = \mathbb{I},$$

where each outcome y occurs with probability

$$\mathbb{P}\{y|\rho\} = \text{tr}(\rho F_y).$$

This framework is sufficiently general to describe the measurement strategies available to an eavesdropper.

D. Maximal Quantum Leakage

Maximal quantum leakage quantifies the largest multiplicative increase in Eve’s probability of correctly guessing a secret variable when she has access to the quantum system. It is defined as [3]

$$\begin{aligned} Q(X \rightarrow A)_\rho &:= \sup_{\{F_y\}_{y \in \mathbb{Y}}} \sup_{Z, \hat{Z}} \log_2 \left(\frac{\mathbb{P}\{Z = \hat{Z}\}}{\max_{z \in \mathbb{Z}} \mathbb{P}\{Z = z\}} \right) \\ &= \sup_{\{F_y\}_{y \in \mathbb{Y}}} \log_2 \left(\sum_{y \in \mathbb{Y}} \max_{x \in \mathbb{X}} \text{tr}(\rho^x F_y) \right). \end{aligned} \quad (1)$$

However, maximal quantum leakage is too permissive for a QKD setting because the supremum is taken over all POVMs, so it does not restrict the disturbance induced by Eve’s measurement. A highly informative measurement may strongly perturb the transmitted state and reveal Eve’s presence, which she intends to avoid.

E. Gentle Measurements and Quantum Leakage

To model an eavesdropper that extracts information while avoiding detection, it is necessary to constrain the admissible measurements by the disturbance they induce on the transmitted quantum states. This is formalized through the notion of gentle measurements [3].

A POVM $\mathcal{F} = \{F_y\}_{y \in \mathbb{Y}}$ is implemented through measurement operators $\mathcal{B} = \{B_y\}_{y \in \mathbb{Y}}$ such that

$$F_y = B_y^\dagger B_y.$$

Given an input state ρ , the post-measurement state conditioned on outcome y is

$$\rho_{B|y} = \frac{B_y \rho B_y^\dagger}{\text{tr}(B_y \rho B_y^\dagger)}.$$

Disturbance is quantified through the trace distance

$$\|\rho - \sigma\|_{\text{tr}} = \frac{1}{2} \text{tr}|\rho - \sigma|.$$

Definition 1 (Gentle Measurements [3]). Let S be a set of quantum states. A POVM $\mathcal{F} = \{F_y\}_{y \in \mathbb{Y}}$ is said to be (α, δ) -weakly gentle on S if there exists an implementation $\mathcal{B} = \{B_y\}_{y \in \mathbb{Y}}$ such that, for every $\rho \in S$,

$$\mathbb{P}\{\|\rho_{B|Y} - \rho\|_{\text{tr}} \leq \alpha, \forall \rho \in S\} \geq 1 - \delta, \quad (2)$$

where the probability is taken with respect to the measurement outcome Y induced by ρ and the chosen implementation. Similarly, a POVM $\mathcal{F} = \{F_y\}_{y \in \mathbb{Y}}$ is said to be (α, δ) -strongly gentle on S if the same condition holds for every possible implementation $\{B_y\}_{y \in \mathbb{Y}}$ of the POVM.

Once the admissible measurements are constrained in this way, one can define *gentle quantum leakage* as the maximum worst-case leakage achievable by an eavesdropper over the class of gentle measurements. Indeed, let X denote Alice's classical random variable, and let Y denote Eve's classical measurement outcome. The corresponding leakage optimization problem can be written as

$$L_{(\alpha, \delta)}(X \rightarrow A)_\rho = \sup_{\{F_y\}_{y \in \mathbb{Y}} \in \mathcal{G}_{(\alpha, \delta)}} I_\infty(X; Y), \quad (3)$$

where $\mathcal{G}_{(\alpha, \delta)}$ denotes the class of admissible weakly gentle measurements [3]. Here, $I_\infty(X; Y)$ is the Sibson mutual information of order infinity between Alice's classical variable X and Eve's measurement outcome Y . It can be understood as the worst-case information gain associated with the measurement outcomes, since it selects the most distinguishable input value for each possible outcome. For a fixed POVM $\mathcal{F} = \{F_y\}_{y \in \mathbb{Y}}$, it is defined as

$$I_\infty(X; Y) := \log_2 \left(\sum_{y \in \mathbb{Y}} \max_{\substack{x \in \mathbb{X}: \\ p_X(x) > 0}} \mathbb{P}[Y = y \mid X = x] \right)$$

$$= \log_2 \left(\sum_{y \in \mathbb{Y}} \max_{x \in \mathbb{X}} \text{tr}(\rho^x F_y) \right). \quad (4)$$

III. NOISE MODEL AND QBER

A. Depolarizing Channel

A central source of noise considered in this work is the depolarizing channel. In dimension d , it is written as

$$D_p(\rho) = p \frac{\mathbb{I}}{d} + (1 - p)\rho. \quad (5)$$

For qubit states, this channel shrinks the Bloch vector and reduces the distinguishability of the encoded states [1]. Therefore, it affects both Bob's decoding performance and Eve's achievable leakage.

From the point of view of this thesis, depolarizing noise plays a double role. On the one hand, it degrades the useful signal available to Eve. On the other hand, it contributes to the overall error rate observed by Alice and Bob, making it necessary to separate natural channel noise from adversarially induced disturbance whenever possible [3], [7].

B. QBER Interpretation

In the BB84 setting, the operational disturbance visible to Alice and Bob is usually expressed through the quantum bit error rate

$$\text{QBER} = \Pr(A \neq B \mid \text{same basis}).$$

A key objective of this thesis is to relate:

$$\alpha \longrightarrow \text{QBER} \longrightarrow L$$

with α being the disturbance Eve introduces in the quantum states when she eavesdrops, and L being the leakage that she obtains by doing so. This relation may not admit a single universal closed-form expression for all attack models. However, even a numerical characterization is already useful, because it allows the leakage analysis to be interpreted in operational QKD language [7].

C. Placement of the QBER Threshold

A practical question is where to place the QBER threshold in the narrative of the thesis. Two interpretations are relevant:

1. A *protocol threshold*, beyond which BB84 would be aborted, as it would no longer be regarded as secure in the usual key-distillation sense;
2. An *attack detectability threshold*, expressing the maximum disturbance Eve can introduce while remaining compatible with realistic operating conditions.

In the first interpretation, the QBER threshold is a *security acceptance criterion*. It determines whether Alice and Bob proceed with the classical post-processing stage or discard the transmission. If the observed QBER is too large, the protocol is aborted. The reason is that a high QBER may indicate either excessive physical noise in the channel or the presence of an eavesdropper. In both cases, the correlations between Alice and Bob are considered too weak to safely extract a secret key through error correction and privacy amplification. In the ideal asymptotic analysis of BB84 with one-way classical post-processing, the QBER threshold is typically set to 11% [5]. Below this value, secret key distillation is feasible; above it, the standard security proof no longer guarantees a positive secret key rate.

This thesis uses the QBER threshold in a second, and more operational sense. Instead of asking only whether Alice and Bob should accept or abort the protocol, we also ask how much disturbance Eve can introduce before her presence becomes detectable. In this interpretation, the QBER threshold defines the admissible region of Eve's attacks. Eve is not allowed to choose arbitrary measurements or cloning transformations, because any attack that produces an excessive QBER would be detected during parameter estimation and would cause the protocol to abort. Thus, the threshold acts as a *detectability constraint*.

IV. ACHIEVABLE LEAKAGE

The main baseline for this section is the gentle quantum leakage framework developed in [3]. This approach makes it possible to study leakage not only as an abstract information-theoretic quantity, but also as a physically attainable benchmark.

A. Asymmetric Quantum Cloning

The analysis of achievable leakage in BB84 naturally starts from the no-cloning theorem: an arbitrary unknown quantum state cannot be copied perfectly. Therefore, any physically realizable attack based on copying must rely on approximate cloning [1], [2].

Approximate cloning produces imperfect copies whose quality can be quantified. In the symmetric case, both copies have the same quality, whereas in the asymmetric case they do not. The latter is especially relevant for eavesdropping, since Eve seeks to optimize the trade-off between the quality of Bob's copy and that of the copy she keeps for herself [3], [8].

Fig. 2 shows the system model of the cloning-based attack, in which Alice's input state is processed by an asymmetric cloner that distributes one imperfect copy to Bob and another to Eve. In the asymmetric setting, the cloner is interpreted as a $1 \rightarrow 2$ quantum channel that acts on Alice's state. One output is sent to Bob, and the other is kept by Eve. The first output determines the disturbance induced on the legitimate communication, while the second one determines the amount of information that Eve can extract. Each

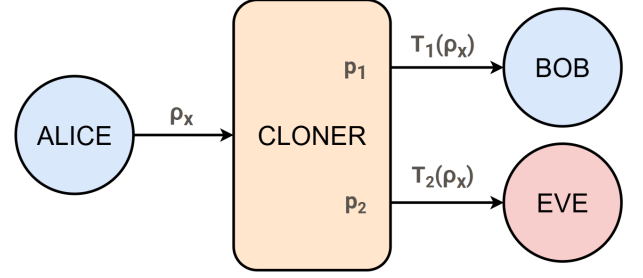


Figure 2. System model for the cloning-based attack.

marginal output channel can be described as a depolarizing channel. More precisely, if ρ denotes Alice's input state, the marginals associated with Bob and Eve can be written as

$$\begin{aligned} T_1(\rho) &= p_1 \rho + (1 - p_1) \frac{\mathbb{I}}{d}, \\ T_2(\rho) &= p_2 \rho + (1 - p_2) \frac{\mathbb{I}}{d}, \end{aligned} \quad (6)$$

where $d = \dim(\mathcal{H})$. In (6), T_1 and T_2 correspond to Bob's and Eve's effective channels [8]. The parameters p_1 and p_2 quantify the quality of the two copies: larger values correspond to outputs that remain closer to the original state, whereas smaller values indicate stronger depolarization. These parameters cannot be chosen arbitrarily. The pair (p_1, p_2) must belong to the physically admissible *quantum cloning region*.

Theorem 1 (Cloning Region [8]). *For all $p_1, p_2 \in [0, 1]$, there exists a quantum cloner T such that*

$$T_i(\rho) = p_i \rho + (1 - p_i) \frac{\mathbb{I}}{d}$$

if and only if

$$\begin{aligned} \|p\|_{\mathcal{Q}} &= \frac{1}{d^2 - 1} \left[\frac{d}{2} (d(|p_1| + |p_2|) \right. \\ &\quad \left. + \sqrt{d^2(|p_1| - |p_2|)^2 + 4|p_1||p_2|}) - (|p_1| + |p_2|) \right] \leq 1 \end{aligned} \quad (7)$$

In (7):

- $\|p\|_{\mathcal{Q}}$ is the Q -norm of the vector p , a dual-norm metric that geometrically characterizes the supporting half-spaces of the admissible cloning region,
- $p = (p_1, p_2)$ is the vector representing the clone-quality parameters (fidelities) of the two marginal outputs,
- $d = \dim(\mathcal{H})$ is the dimension of the quantum system.

The maximum achievable cloning fidelities, where the optimal performance limits dictate that one clone's

quality cannot be improved without degrading the other, are obtained by establishing the equality $\|p\|_Q = 1$. This condition defines the optimal asymmetric cloning trade-off curve, shown in Fig. 3.

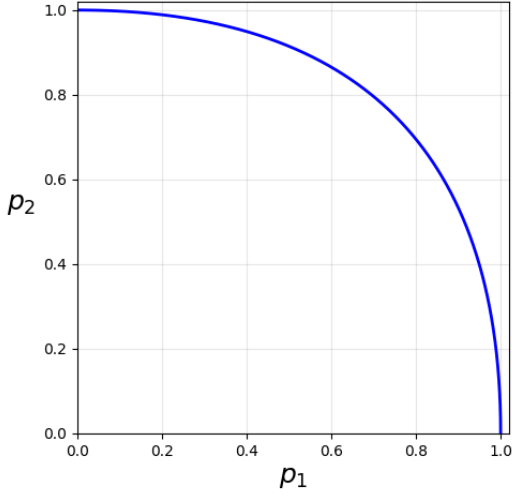


Figure 3. Optimal values of p_1 and p_2 for asymmetric cloning [8].

The values bounded beneath the curve represent the physically admissible states within the cloning region. The boundary curve itself is achieved when the inequality in Theorem 1 is satisfied with equality. As a result, Bob's copy quality and Eve's copy quality cannot be optimized independently: improving Eve's copy generally worsens Bob's copy, and vice versa. This induces a trade-off between information leakage and disturbance.

A cloning-based lower bound is obtained by restricting p_1 so that Bob's copy satisfies the required disturbance constraint, and then maximizing p_2 over all admissible pairs (p_1, p_2) in the cloning region. The resulting value determines the best Eve-side copy compatible with the chosen disturbance level, and thus yields a lower bound on the achievable leakage [3]. By using this strategy, Eve's clone has $\log_2((1 - p_2) + p_2 2^{Q(X \rightarrow A)})$ bits of information.

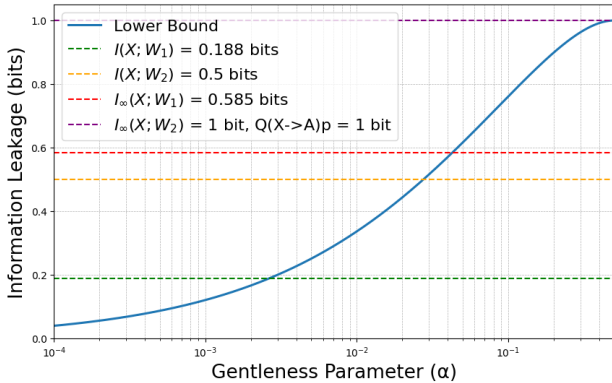


Figure 4. Lower bound derived from Asymmetric Quantum Cloning [3].

Fig. 4 shows the cloning-based lower bound on Eve's achievable information leakage as a function of the gen-

teness parameter α defined in (2). The figure also includes reference leakage values for two standard, non-cloning-based projective measurement strategies:

- **Strategy W_1 :** Eve performs a projective measurement choosing between the Z and X bases uniformly at random.
- **Strategy W_2 :** Eve performs a projective measurement consistently in a single fixed basis (either Z or X).

In the figure, $I(X; W)$ and $I_\infty(X; W)$ denote the standard Shannon mutual information and the Sibson mutual information (of order ∞) for these respective strategies. As α increases, the lower bound monotonically rises until it saturates at 1 bit, eventually matching the operational leakage limit $Q(X \rightarrow A)$. Since $I_\infty(X; W)$ captures a worst-case leakage scenario, it is the more conservative and therefore more suitable quantity for security analysis.

B. Achievable Leakage Under Noise

So far, the cloning-based lower bound and the POVM-based attacks have been considered in the ideal noiseless setting. However, the main focus of this work is to understand how depolarizing noise affects the amount of information that an eavesdropper can extract.

As the depolarization increases, the transmitted states become closer to the maximally mixed state, their distinguishability decreases, and therefore the amount of useful information available to Eve is reduced.

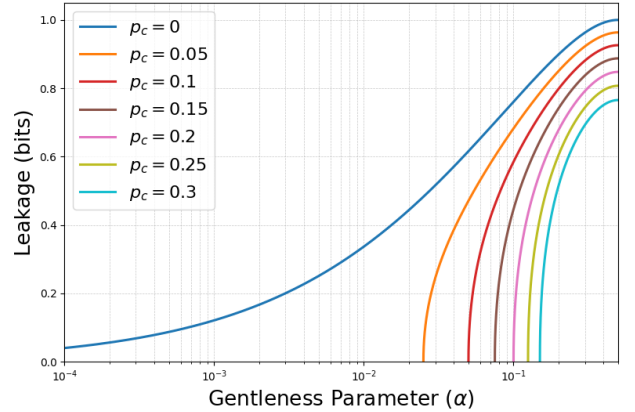


Figure 5. Effect of noise on information leakage.

Fig. 5 illustrates the effect of depolarizing noise on the cloning-based lower bound for Eve's achievable leakage. As the depolarizing coefficient p_c increases, the curves are shifted downwards, meaning that for the same gentleness parameter α Eve obtains less information. This confirms that noise reduces the distinguishability of the transmitted states and therefore weakens Eve's effective leakage, although the quantitative impact depends on the specific attack strategy considered.

V. QKD PARAMETER MAPPING

We now establish direct connections between our channel parameters (p_1, p_2) and standard QKD figures of merit, such as the Quantum Bit Error Rate (QBER), the gentleness parameter (α) , and Eve's information leakage. Defining these explicit relationships not only provides a more intuitive, physical understanding of the system's trade-offs, but also allows us to readily translate bounds on cloning fidelity into actionable security thresholds and performance guarantees for the protocol.

A. Ideal scenario

Consider the system presented in Sec. II, with no channel noise and where Eve implements an attack using an asymmetric quantum cloner.

Relation between QBER and p_1

In this scenario, Bob receives the first clone of Alice's qubit, which is described by the depolarized state

$$T_1(\rho) = p_1\rho + (1 - p_1)\frac{I}{2}.$$

In sifted BB84, Bob measures in the correct basis. Therefore, for a given transmitted state ρ_x , the error event is the projection onto the orthogonal state in the same basis, namely $I - \rho_x$. Hence, the conditional bit error probability is

$$\begin{aligned} \text{QBER}(x) &= \text{tr}[(I - \rho_x) T_1(\rho_x)] \\ &= \text{tr}\left[(I - \rho_x) \left(p_1\rho_x + (1 - p_1)\frac{I}{2}\right)\right] \\ &= p_1 \text{tr}[(I - \rho_x)\rho_x] + \frac{1 - p_1}{2} \text{tr}(I - \rho_x). \end{aligned}$$

Since ρ_x is a rank-one projector, we have $\rho_x^2 = \rho_x$. And, therefore,

$$(I - \rho_x)\rho_x = \rho_x - \rho_x^2 = 0.$$

Also, because we are in dimension 2,

$$\text{tr}(I - \rho_x) = 2 - 1 = 1.$$

Thus,

$$\text{QBER} = \frac{1 - p_1}{2}. \quad (8)$$

Relation between α and p_1

The actual disturbance induced on the state ρ_x is

$$\Delta_x := \|T_1(\rho_x) - \rho_x\|_{\text{tr}}.$$

Using the definition of T_1 (see (6)) and taking the trace norm, we obtain

$$\begin{aligned} \Delta_x &= (1 - p_1) \left\| \frac{I}{2} - \rho_x \right\|_{\text{tr}} \\ &= (1 - p_1) \frac{1}{2}. \end{aligned}$$

If the measurement is required to be α -gentle, then the actual disturbance must satisfy $\Delta_x \leq \alpha$, so

$$\frac{1 - p_1}{2} \leq \alpha. \quad (9)$$

Relation between QBER and α

From (8) and (9), we can infer that in this model

$$\text{QBER} = \Delta_x \leq \alpha, \quad (10)$$

with equality if the gentleness constraint is active.

Relation between leakage and p_2

We have seen before that using asymmetric cloning, Eve's copy contains

$$\begin{aligned} L_{\text{LB}} &= \log_2 \left((1 - p_2) + p_2 2^{Q(X \rightarrow A)_\rho} \right) \\ &= \log_2((1 - p_2) + 2p_2) \\ &= \log_2(1 + p_2) \end{aligned}$$

bits of information, given that for the BB84 ensemble the maximal quantum leakage in the ideal case is $Q(X \rightarrow A)_\rho = 1$. Therefore,

$$L_{\text{LB}}(p_2) = \log_2(1 + p_2). \quad (11)$$

This is a lower bound, not an exact universal expression for the leakage, but it will be used as the baseline for our estimations.

B. Noisy scenario: additional depolarizing channel

We now introduce an additional physical depolarizing channel with coefficient p_c between Alice and the cloner:

$$\mathcal{D}_{p_c}(\rho) = (1 - p_c)\rho + p_c \frac{I}{2}.$$

Effective parameters p'_1 and p'_2

We assume that this extra noise acts after the cloning stage on each branch. Then, the effective channel on branch i becomes

$$\begin{aligned} T'_i(\rho) &= T_i(\mathcal{D}_{p_c}(\rho)) \\ &= (1 - p_c) \left(p_i\rho + (1 - p_i)\frac{I}{2} \right) + p_c \frac{I}{2} \\ &= (1 - p_c)p_i\rho + ((1 - p_c)(1 - p_i) + p_c) \frac{I}{2}. \end{aligned}$$

Observe that

$$(1 - p_c)(1 - p_i) + p_c = 1 - (1 - p_c)p_i.$$

Therefore,

$$T'_i(\rho) = p'_i\rho + (1 - p'_i)\frac{I}{2},$$

with

$$p'_i = (1 - p_c)p_i, \quad i \in \{1, 2\}. \quad (12)$$

QBER under additional depolarizing noise

Replacing p_1 by p'_1 in (8), we obtain

$$\text{QBER}' = \frac{1 - (1 - p_c)p_1}{2}. \quad (13)$$

This shows explicitly how the additional depolarizing noise increases the observed QBER.

Relation between α and p'_1

After replacing p_1 by p'_1 in (9), the disturbance is now

$$\frac{1 - (1 - p_c)p_1}{2} \leq \alpha \quad (14)$$

when we impose α -gentleness. Under depolarizing noise, we can see that (10) still holds.

Leakage under additional depolarizing noise

Replacing p_2 by $p'_2 = (1 - p_c)p_2$ in the ideal leakage lower bound (11), yields

$$L'_{\text{LB}} = \log_2(1 + (1 - p_c)p_2). \quad (15)$$

Hence, the additional depolarizing noise reduces the information leakage from Alice to Eve.

VI. IMPACT OF THE DEPOLARIZING CHANNEL

We have seen that the depolarizing channel reduces the information leakage to Eve, but also deteriorates the information shared between Alice and Bob. This raises the following question: does the presence of depolarizing noise harm Eve more than it harms the legitimate parties Alice and Bob?

To analyze this, we will consider the asymmetric cloning model used previously. After basis sifting, Alice and Bob keep only the events in which they used the same measurement basis. At this point, the quantum transmission can be represented as an effective classical binary channel between Alice's bit and Bob's bit. In the present model, the errors are symmetric, so this channel can be treated as a binary symmetric channel (BSC), whose bit-flip probability is precisely the observed QBER (from (13)) [9]. The capacity of this channel is therefore

$$C_{AB}(p_c) = 1 - h_2\left(\frac{1 - (1 - p_c)p_1}{2}\right), \quad (16)$$

where $h_2(\cdot)$ is the binary entropy function [10]. In this context, $C_{AB}(p_c)$ represents the maximum useful information per sifted bit that Alice and Bob can reliably retain after ideal classical error correction. Therefore, it provides an upper bound on the effective Alice–Bob information available before privacy amplification.

On Eve's side, using the leakage expression adopted in the cloning model, the leakage becomes

$$L_E(p_c) = \log_2(1 + (1 - p_c)p_2). \quad (17)$$

A. Information advantage

To determine which side is affected the most, Fig. 6 shows the information advantage, defined as

$$\Delta_{\text{info}}(p_c) = C_{AB}(p_c) - L_E(p_c). \quad (18)$$

Positive values of $\Delta_{\text{info}}(p_c)$ mean that Alice and Bob retain more useful information than Eve obtains, according to the metrics used in this model. Conversely, negative values indicate that Eve has more information than the Alice–Bob channel can reliably support.

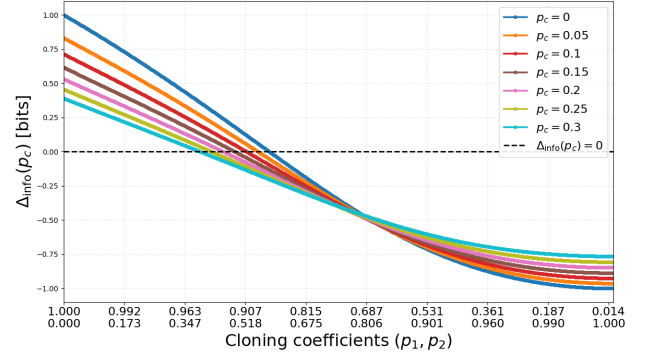


Figure 6. Information advantage for different cloning coefficients.

Several qualitative observations can be drawn from the figure. First, the comparison is deliberately conservative from Alice and Bob's perspective. The quantity $C_{AB}(p_c)$ represents the effective useful information shared by Alice and Bob through the noisy channel, whereas $L_E(p_c)$ is computed from Eve's Sibson mutual information, which captures a worst-case information leakage criterion. Therefore, the plotted quantity

$$\Delta_{\text{info}}(p_c) = C_{AB}(p_c) - \mathcal{L}_E(p_c)$$

does not compare two identical operational quantities. This makes the analysis biased towards detecting potentially dangerous regimes, which is appropriate when the goal is to assess security under conservative assumptions.

Second, the figure shows that Eve can obtain an information advantage for a significant part of the cloning boundary. This is especially the case in regions where the cloning asymmetry favors Eve, corresponding to larger values of p_2 and smaller values of p_1 . However, these points are also associated with a stronger disturbance on Bob's received states. Hence, they cannot be interpreted independently of the QBER that they would induce in the protocol.

Third, increasing depolarizing noise appears to flatten the information-advantage curve. This behavior suggests that depolarizing noise does not simply benefit one side uniformly. Rather, it compresses the information gap between Alice–Bob and Eve.

B. QBER threshold

The QBER threshold therefore plays a central role in restricting Eve's admissible strategies. Using the Shor–

Preskill [5] threshold of approximately 11% as a reference, we identify the portion of the cloning region in which Eve can extract information while remaining below the disturbance level tolerated by the protocol.

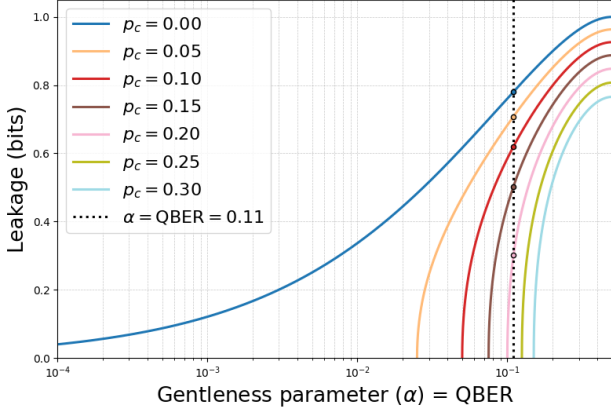


Figure 7. Eve's leakage under the QBER constraint for different depolarizing-noise levels.

Fig. 7 shows Eve's leakage obtained from the approximate cloning strategy for different depolarizing-noise levels p_c . The vertical dotted line marks the Shor–Preskill QBER threshold

$$QBER_{th} = 0.11,$$

and the highlighted points indicate the maximum leakage achieved within the asymmetric-cloning lower-bound model while remaining below this threshold. These points therefore represent the most favorable admissible strategies for Eve under the constraint that the protocol is not aborted.

The figure illustrates that the QBER threshold directly limits Eve's operational region. A larger accepted QBER would allow Eve to operate at higher values of α , and therefore to obtain more leakage before being detected. Conversely, a stricter QBER threshold would reduce the admissible region and limit the maximum leakage available to Eve. It can also be observed that, for a fixed QBER threshold, the presence of depolarizing noise significantly reduces Eve's obtainable leakage. This suggests that background channel noise reduces Eve's effective information extraction, while also consuming part of the QBER budget allowed by the protocol.

To summarize, Fig. 7 quantifies the maximum leakage that Eve can obtain while remaining below the 11% QBER threshold. However, this value alone does not determine whether the communication should be considered secure, since it only describes Eve's side of the trade-off. To assess the security implications of these admissible leakage values, they must be compared with the effective information shared by Alice and Bob. This motivates the following analysis in terms of the information advantage Δ_{info} .

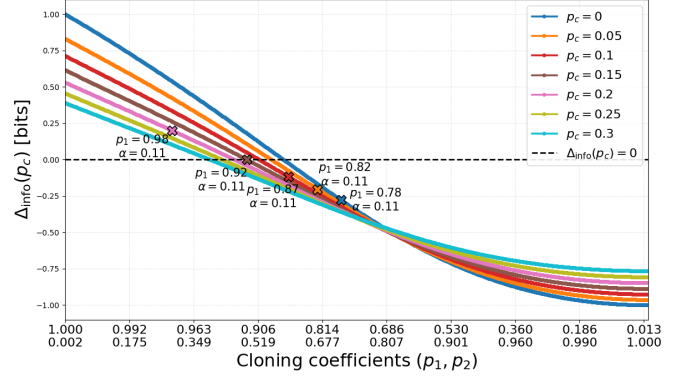


Figure 8. Best information advantage obtainable by Eve at the QBER constraint for different depolarizing-noise levels.

Fig. 8 shows the information advantage curve with the operating points identified in Fig. 7. That is, for each value of the depolarizing-noise parameter p_c , the marked point represents the information advantage associated with Eve's maximum leakage while still remaining below the QBER threshold. Therefore, these points represent the most favorable undetected strategies for Eve under the imposed QBER constraint. Two main observations can be made:

1. For low depolarizing-noise levels, Eve can still obtain an information advantage over Alice and Bob.
2. As the depolarizing noise increases, the information advantage shifts progressively in favor of Alice and Bob.

It is also worth noting that, if we require

$$\Delta_{info} \geq 0$$

as a condition for secure communication, then the standard QBER threshold may be too permissive for the present leakage-based analysis. Indeed, Eve may remain below the usual Shor–Preskill threshold while still obtaining a worst-case leakage larger than the effective information shared by Alice and Bob.

Our analysis allows us to rigorously establish the QBER threshold, for which Eve achieves no information advantage provided that the parameter of the depolarizing noise is known. Specifically, this threshold can be chosen by imposing the worst-case boundary condition

$$\Delta_{info} = 0.$$

In this way, the protocol only accepts disturbances for which Alice and Bob retain a non-negative information advantage over Eve. The most permissive QBER threshold is illustrated in Fig. 9. Observe that the Shor–Preskill threshold guarantees a positive information advantage only for noise parameters above 0.15.

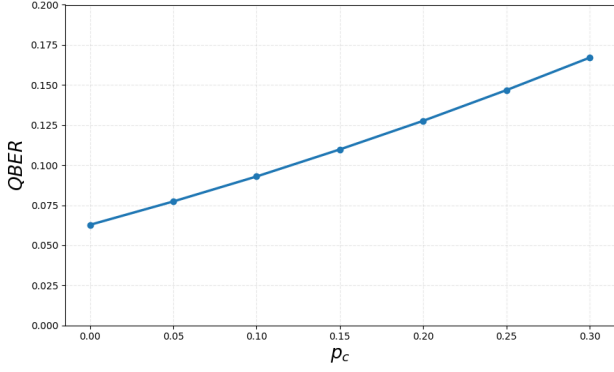


Figure 9. Most permissive QBER threshold that guarantees information advantage for different noise levels.

Alternatively, if the average depolarizing noise of the channel is known beforehand, the QBER threshold can be used as a direct constraint on Eve’s leakage, independently of the information-advantage criterion. Under this approach, one fixes a maximum tolerable leakage $\mathcal{L}_E^{\max}$ and chooses the QBER threshold so that every cloning strategy remaining below the threshold also satisfies

$$\mathcal{L}_E(p_2, p_c) \leq \mathcal{L}_E^{\max}.$$

This provides an operational way of calibrating the accepted disturbance level: the threshold is no longer interpreted only as a protocol-abort condition, but also as a mechanism to limit the amount of information that Eve can obtain while remaining undetected.

VII. CONCLUSION

The results suggest that the QBER threshold can be interpreted not only as a protocol-abort criterion, but also as an operational constraint on Eve’s admissible leakage. By relating the cloning parameters to the induced QBER, Eve’s leakage, and the information advantage, the performed analysis provides a useful way to evaluate whether an attack can remain undetected while still being informationally relevant.

In particular, the analysis shows that depolarizing noise has a non-trivial effect. Although it degrades the useful information shared by Alice and Bob, it also reduces the quality of Eve’s copy and consumes part of the QBER budget available to her. As a consequence, for a fixed QBER threshold, the presence of noise for may be beneficial against eavesdropping.

Future work could incorporate error correction and privacy amplification explicitly, so that the leakage-based analysis can be connected to final secret-key rates. It would also be useful to extend the model to finite-size effects, more realistic noise sources, and alternative eavesdropping strategies beyond the asymmetric cloning bound considered here.

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